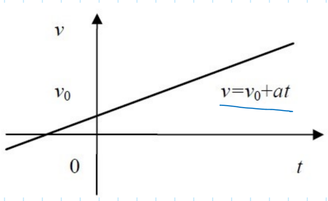
Ligjerata 2dh

Lëvizje e njëtrajtshme e ndryshueshme drejtvizore

$\vec{a}$ = konst. ose a = konst.

$$a = \frac{dv}{dt} = \frac{d}{dt}\left(\frac{ds}{dt}\right) = \frac{d^2s}{dt^2}$$



$$a = \frac{dv}{dt} \Big/ \cdot dt$$

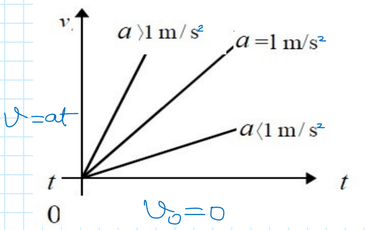
$$dv = a\,dt \Big/ \int$$

$$\int_{v_0}^{v} dv = a\int_0^t dt$$

$$v\Big|_{v_0}^{v} = a(t\Big|_0^t)$$

$$v - v_0 = at$$

$$\boxed{v = v_0 + at}$$



$$v = \frac{ds}{dt} \Big/ \cdot dt \rightarrow ds = v\,dt$$

$$ds = (v_0 + at)\,dt$$

$$ds = v_0\,dt + at\,dt \Big/ \int$$

$$\int_0^s ds = v_0\int_0^t dt + a\int_0^t t\,dt$$

$$\boxed{s = v_0 t + \frac{at^2}{2}}$$

$$\int dx = x + C$$

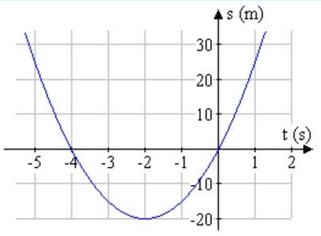
$$\int x^m dx = \frac{x^{m+1}}{m+1} + C$$

$$m = 1 \quad = \frac{x^2}{2} + C$$

$$v = v_0 + at \rightarrow v - v_0 = at \Big/ : a$$

$$t = \frac{v - v_0}{a}$$

$$s = v_0 t + \frac{at^2}{2}$$



$$s = v_0\left(\frac{v - v_0}{a}\right) + \frac{a}{2}\left(\frac{v - v_0}{a}\right)^2$$

$$s = \frac{v_0 v - v_0^2}{a} + \frac{v^2 - 2vv_0 + v_0^2}{2a} = \frac{v_0 v - v_0^2}{a} + \frac{v^2 - 2vv_0 + v_0^2}{2a}$$

$$s = \frac{2v_0 v - 2v_0^2 + v^2 - 2vv_0 + v_0^2}{2a}$$

$$s = \frac{v^2 - v_0^2}{2a} \Big/ \cdot 2a \rightarrow 2as = v^2 - v_0^2$$

$$\boxed{v^2 = v_0^2 + 2as}$$

$a = konst > 0$ lëvizja është njëtrajtësisht e nxituar

$a = konst < 0$ — " — — " — e ngadalsuar.

$$\left.\begin{aligned} v &= v_0 + at \\ s &= v_0 t + \frac{1}{2}at^2 \end{aligned}\right\} \quad \begin{aligned} s &= h \\ a &= g \end{aligned}$$

$$\left.\begin{aligned} v &= v_0 + gt \\ h &= v_0 t + \frac{1}{2}gt^2 \end{aligned}\right\} \quad \begin{aligned} v_0 = 0 &\rightarrow \boxed{v = gt} \\ v_0 = 0 &\rightarrow \boxed{h = \frac{1}{2}gt^2 \quad v^2 = 2gh} \end{aligned}$$

$$v - v_0 = gt \Big/ : g \rightarrow t = \frac{v - v_0}{g}$$

$$h = v_0\left(\frac{v - v_0}{g}\right) + \frac{1}{2}g\left(\frac{v - v_0}{g}\right)^2$$

$$g = 9{,}81\ \frac{m}{s^2}$$

$$h = \frac{v_0 v - v_0^2}{g} + \frac{v^2 - 2vv_0 + v_0^2}{2g} = \frac{2vv_0 - 2v_0^2 + v^2 - 2vv_0 + v_0^2}{2g}$$

$$h = \frac{v^2 - v_0^2}{2g} \Rightarrow 2gh = v^2 - v_0^2$$

$$\boxed{v^2 = v_0^2 + 2gh}$$